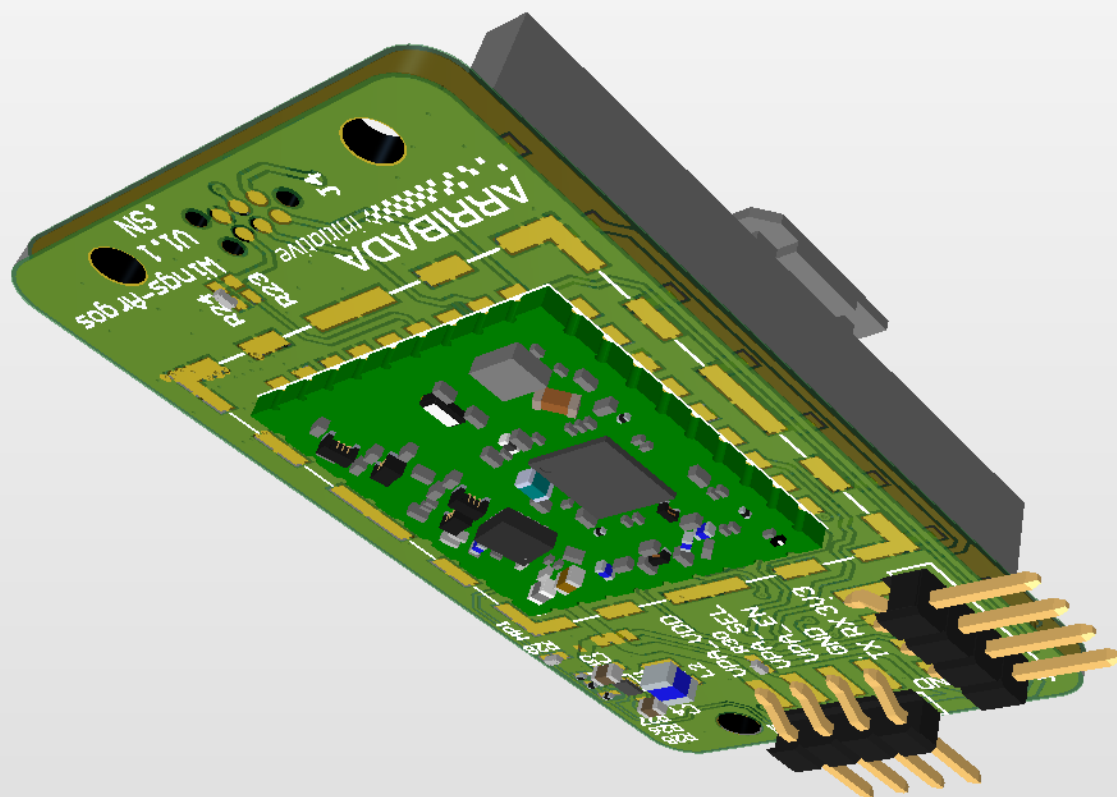




HorizonLink-Wings_Breakout

smd-wings

Feather wing for the ARGOS-SMD module. This breakout board can be used with a Feather board (from Adafruit) to connect the ARGOS-SMD module. It exposes the ARGOS-SMD module pins to the Feather board and includes a specific LDO to power the RF amplifier.

Table of contents

- Sheet 1: Project overview (this page)
- Sheet 2: SMD_Breakout
- Device Sheets:
 - Sheet 3: TPS63901_BuckBoost_lowIq_3V3_400mA

Legend

- General comment such as function title, configuration, ...
- Text to be added to silkscreen.
- Warning text.



OPEN HARDWARE LICENSE AGREEMENT

The Open Hardware License Agreement (“Agreement”) is a binding legal contract between you (“You”) and the creators of this design, as represented by the Arribada Initiative, under the terms of the GNU General Public License version 3 (“GPLv3”). This license governs the use, modification, and distribution of the open-design PCB and any associated design files or materials (collectively referred to as the “Reference Material”).

By accessing, using, or modifying the Reference Material, You agree to be bound by the terms and conditions of the GPLv3. If you do not agree to these terms, do not use the Reference Material. Your use of the Reference Material constitutes acceptance of this Agreement.

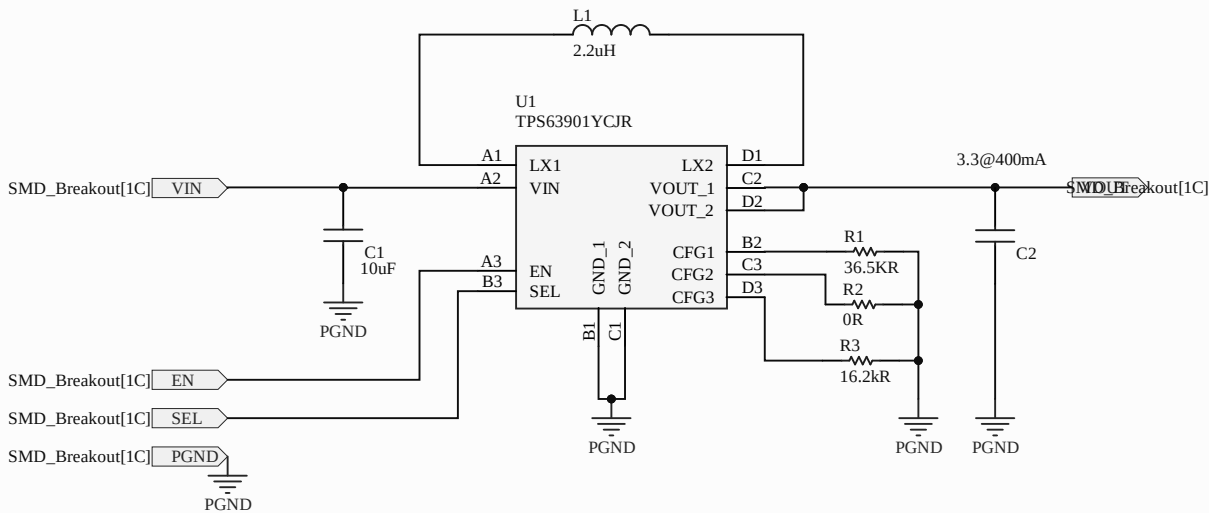
The complete GNU General Public License version 3 can be found at www.gnu.org/licenses/gpl-3.0.html.

For more details about this open-design PCB project and its licensing, please visit the Arribada Initiative website or contact us directly.

smd-wings



Title HorizonLink-Wings_Breakout			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth		
Size: A4	Reference: smd-wings	Revision: B 1.1			
Date: 06/08/2025	Time: 15:22:21	Sheet 1 of 3			
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\featherwings-argos-smd-hw\Project_Overview.SchDoc					



Title HorizonLink-Wings_Breakout				Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth		 initiative
Size: A4	Reference: smd-wings	Revision: B	1.1			
Date: 06/08/2025	Time: 15:22:21	Sheet 3	of 3			
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\TPS63901_BuckBoost_lowIq_3V3_400mA						

Board Stack Report

Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
1		Top Overlay		0mil	
2		Top Solder	SM-001	1mil	4
3		Top Surface Finish	PbSn	0.787mil	
4		Top Layer	CF-004	1.378mil	
5		Dielectric 1	Core-043	59mil	4.3
6		Bottom Layer	CF-004	1.378mil	
7		Bottom Surface Finish	PbSn	0.787mil	
8		Bottom Solder	SM-001	1mil	4
9		Bottom Overlay		0mil	
	Height : 65.331mil				

Comment	Description	Designator	Footprint	LibRef	Quantity
GRM155R71C104KA88D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.1 uF 16 VDC 10% 0402 X7R	C1	0402	GRM155R71C104KA88D	1
CC0402BRNPO9BN3R6	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50 V 3.6pF COG 0402 Tol 0.1pF	C3	CAPC1005X55N	CC0402BRNPO9BN3R6	1
GRM188R61A106MAALJ	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 uF 10 VDC 20% 0603 X5R	C4	CAPC1608X90N	GRM188R61A106MAALJ	1
GRM188R61A226ME15D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 22 uF 10 VDC 20% 0603 X5R	C5	GRM18x	GRM188R61A226ME15D	1
MBR120LSFT3G	Schottky Diodes & Rectifiers 1A 20V	D1, D2	SODFL3616X98N	MBR120LSFT3G	2
20279-001E-01	CON IPEX MHF SMT RECEPTACLE RF DC - 6GHz VSWR 1.3 max at 0.1-3GHz VSWR 1.5 max. at 3-6GHz	J1	MHF SMT REC	20279-001E-01	1
SSM-112-F-SV-P-TR	Headers & Wire Housings Surface Mount, 12pos, PCB Socket Strips, .100" pitch	J2	SAMTEC_SSM-112-F-SV-P-TR	SSM-112-F-SV-P-TR	1
SSM-116-F-SV-P-TR	Headers & Wire Housings Surface Mount, 16pos, PCB Socket Strips, .100" pitch	J3	SAMTEC_SSM-116-F-SV-P-TR	SSM-116-F-SV-P-TR	1
TC2030-NL	CON PLUG OF NAILS 6 PIN FOOTPRINT NO LEGS	J4	TC2030NL	TC2030-NL	1
M20-8890445R	Connector	J5, J6	M208890445R	M20-8890445R	2
PRT-14417	QWIC .5T CONNECTOR- SMD 4-PIN	J7	SPARKFUN_PRT-14417	PRT-14417	1
LQW18AN18NG8ZD	Inductor Wirewound, 18 nH, 75 Milliohm, ± 2%, 1.4 A, 0603 [1608 Metric]	L1	LQW18AN10NG80D	LQW18AN18NG8ZD	1
DfE252012F-2R2M-P2	Inductor Power Chip Shielded Wirewound 2.2uH 20% 1MHz Metal 2.3A 0.082 Ohm DCR 1008 T/R	L2	DfE2HCAH1R0MJDL	DfE252012F-2R2M-P2	1
DMG3415U-7	MOSFET (P-Channel) 16V Bnh FET 8Vgss - 12A 0.65W	Q1	SOT96P240X120-3N	DMG3415U-7	1
BSS138BKW,115	MOSFET (N-Channel)	Q2	SOT65P210X110-3N	BSS138BKW,115	1
RC0805JR07100KL	Thick Film Resistors - SMD 100 kOhms 125 mW 0805 5%	R1	RESC2012X60N	RC0805JR07100KL	1
0R	Thick Film Resistors - SMD 0402 0.063 W 0 Ohm Jumper	F5, F6, R17, R18, R24	RESC1005X40N	CRCW04020000020ED	5
CRCW020136K5FNED	Thick Film Resistors - SMD 36.5Kohm 1% 200ppm	R25	RESC0603X28N	CRCW020136K5FNED	1
RC0201FR-070RL	Thick Film Resistors - SMD 0 Ohms 50mW 0201 1%	R26	RESC0603X26N	RC0201FR-070RL	1
ERJ-1GNF1622C	Thick Film Resistors - SMD 0201 16.2Kohm 1% Halogen Free AEC-Q200	R27	ERJ1GN_(0201)	ERJ-1GNF1622C	1
CRCW040210K0JNEDC	Thick Film Resistors - SMD 1/16watt 10Kohms 5% Commercial Use	R28, R30	RESC1005X40N	CRCW040210K0JNEDC	2
stm32w-SMD	Arribada SMD module Argos communication with PA GR5504	SMD1	stm32w-SMD_noLegs	stm32w-SMD	1
PTS820J2x	SW SPST PB 50MA 12V SMT CK variant	SW1	TL1015	PTS820J2x	1
TPS63901YCIJR	Integrated Circuit	U1	BGA12C35P3X4_111X146X35	TPS63901YCIJR	1